Cloaking experiments with Nassar-type metamaterials
(notes)

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1 Simple homogenization schema of Nassar-type unit cells

1.1 Transformation elasticity and the BGM gauge

An elastic cloak is an annular coating $\{r_i < \|\mathbf{x}\| < r_c\}$ that renders a void $\{\|\mathbf{x}\| < r_i\}$ invisible to elastic waves. The constitutive properties of the coating follow from a blow-up coordinate transformation $\mathbf{x} = \boldsymbol{\phi}(\mathbf{X})$ that maps the origin to the circle of radius r_i [BGM09].

Under the Brun–Guenneau–Movchan (BGM) gauge, the transformed elasticity tensor and mass density are [NCH18]

$$c_{ijkl} = \frac{F_{jm}\, F_{ln}\, C_{imkn}}{J}\,, \qquad \rho = \frac{R}{J}\,, \tag{1}$$

where $\mathbf{F} = \partial\mathbf{x}/\partial\mathbf{X}$ is the deformation gradient, $J = \det \mathbf{F}$, $\mathbf{C}$ and R are the background stiffness and density. The transformed tensor $\mathbf{c}$ breaks the minor symmetries ($c_{ijkl} \neq c_{jikl}$) while preserving major symmetry ($c_{ijkl} = c_{klij}$). A material realising such a tensor is necessarily *polar* (it resists rotations via body torques) and *degenerate* (it admits a stressless collapse mechanism) [NCH18].

For the circular blow-up with an isotropic background (λ, μ), the effective stiffness in the local polar basis $(\mathbf{m}, \mathbf{n})$ = (radial, tangential) reads

$$\mathbf{c}_{\text{eff}} = \begin{bmatrix} (2\mu{+}\lambda)\, f & \lambda & 0 & 0 \\ \lambda & (2\mu{+}\lambda)/f & 0 & 0 \\ 0 & 0 & \mu/f & \mu \\ 0 & 0 & \mu & \mu f \end{bmatrix}, \qquad f = \frac{\|\mathbf{x}\| - r_i}{\|\mathbf{x}\|}\,, \tag{2}$$

in augmented Voigt ordering $[\sigma_{11}, \sigma_{22}, \sigma_{12}, \sigma_{21}]$. The effective density is position-dependent:

$$\rho_{\text{eff}} = \frac{r_c^2}{(r_c - r_i)^2}\, f\, R\,. \tag{3}$$

Near the inner boundary ($r \to r_i$, $f \to 0$) the radial stiffness and density vanish while the tangential stiffness diverges — perfect cloaking requires this singular behaviour, which can only be approximated by a finite lattice.

1.2 The Nassar degenerate polar lattice

Nassar et al. [NCH18] propose a lattice whose unit cell contains a rigid rectangular mass with two diagonal springs of constant α, one vertical spring of constant β, and a torsion spring of constant κ. The lattice is rectangular with cell dimensions a (radial) and b (tangential), tilted at angle θ to the radial direction (Fig. 1).

The lattice's effective Hooke's law in augmented Voigt is

$$\mathbf{c}_{\text{lat}} = \begin{bmatrix} r\,c^2\alpha & cs\alpha & 0 & 0 \\ cs\alpha & (\alpha s^2{+}2\beta)/r & 0 & 0 \\ 0 & 0 & c^2\alpha\kappa/\Delta & r\,cs\alpha\kappa/\Delta \\ 0 & 0 & r\,cs\alpha\kappa/\Delta & r^2s^2\alpha\kappa/\Delta \end{bmatrix}, \tag{4}$$

where $c = \cos\theta$, $s = \sin\theta$, $r = a/b$ is the cell aspect ratio, and $\Delta = \alpha(c - rs)^2 + r\kappa$.

1.3 Parameter identification

Matching (4) to (2) yields the design parameters that make the lattice reproduce the transformation cloaking tensor exactly. The global parameters (position-independent) are

$$\theta = \arctan\sqrt{\frac{\lambda}{2\mu+\lambda}}\,, \quad \alpha = 2(\mu+\lambda)\sqrt{\frac{\lambda}{2\mu+\lambda}}\,, \quad \beta = \frac{(2\mu+\lambda)^2 - \lambda^2}{2\sqrt{\lambda}\,\sqrt{2\mu+\lambda}}\,, \tag{5}$$

and the position-dependent parameters (through f) are

$$\frac{a}{b} = f\sqrt{\frac{2\mu+\lambda}{\lambda}}\,, \qquad \kappa = \frac{\lambda\mu}{\lambda-\mu}\,\frac{(1-f)^2}{f}\,. \tag{6}$$

Stability requires $\mu \le \lambda$ so that $\kappa > 0$. In the special case $\lambda = \mu$, we obtain $\kappa \to \infty$ (the lattice enforces minor symmetry) and $\theta = \pi/6$.

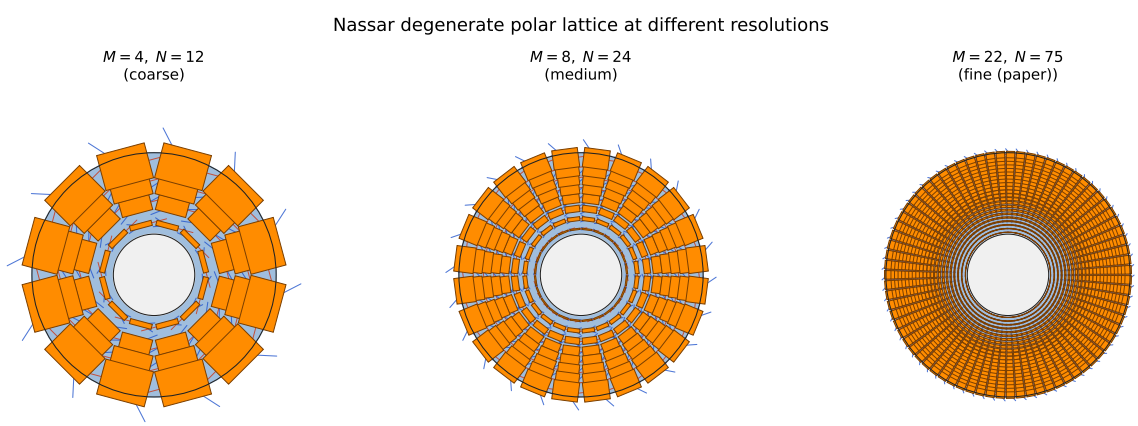

Figure 1: Nassar degenerate polar lattice at three resolutions. Each cell contains a rigid mass (orange), two diagonal springs α (blue), and one vertical spring β (red). The innermost ring of cells cannot reach $r = r_i$ exactly; cloaking becomes ideal only in the limit $M, N \to \infty$.

1.4 Homogenized cell model

Instead of modelling the discrete lattice, we discretise the cloak annulus into a regular Cartesian grid of $n_x \times n_y$ square cells (Fig. 2). Each cell whose centre lies in $\{r_i < r < r_c\}$ is assigned a piecewise-constant effective stiffness $\mathbf{c}_{\text{lat}}$ computed from (4) with parameters from (5)–(6) evaluated at the cell centre. The

density is set to (3). Cells outside the annulus retain the background material; cells inside the void are removed from the mesh.

Since the Nassar parameters are expressed in the local polar frame $(\mathbf{m}, \mathbf{n})$, the stiffness tensor must be rotated to the global Cartesian frame (x, y) by the standard fourth-order rotation:

$$c^{\text{cart}}_{ijkl} = R_{ia}\, R_{jb}\, R_{kc}\, R_{ld}\, c^{\text{polar}}_{abcd}\,, \qquad \mathbf{R}(\varphi) = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}, \tag{7}$$

where $\varphi = \operatorname{atan2}(y, x)$ is the polar angle of the cell centre.

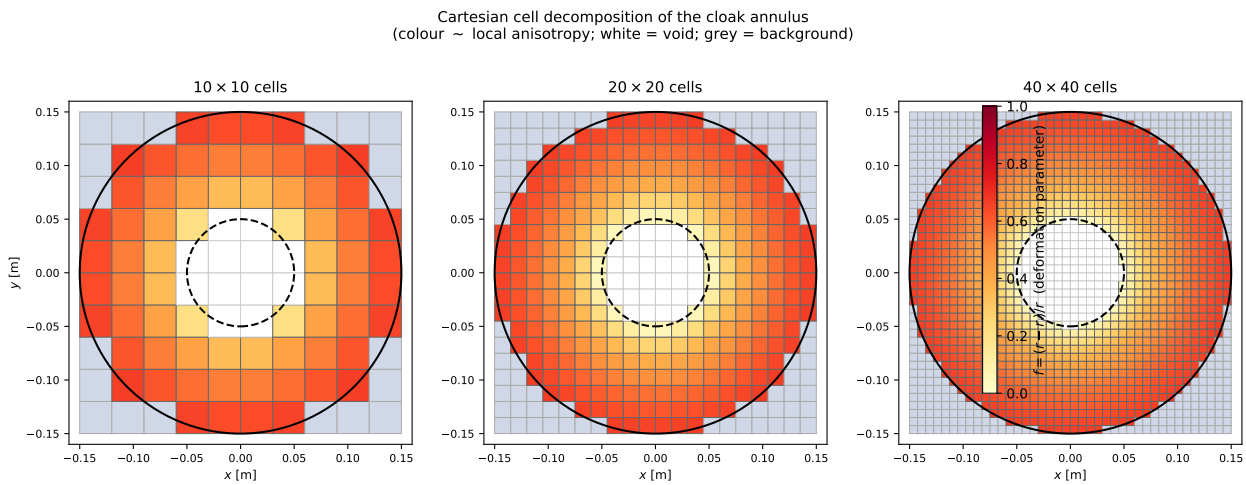

Figure 2: Cartesian cell decomposition of the cloak annulus at three grid resolutions. Colour encodes the deformation parameter $f = (r - r_i)/r$, which controls the local anisotropy; $f \to 0$ near r_i (near-singular) and $f \to (r_c - r_i)/r_c$ near r_c (close to background). White cells lie inside the void; grey cells are background. Dashed/solid circles mark $r = r_i$ and $r = r_c$.

2 Circular cloaking of plane waves

2.1 Simulation setup

We consider a rectangular domain of side $4r_c = 0.6\,\text{m}$ with PML absorbing layers on all four boundaries. A circular void of radius $r_i = 50\,\text{mm}$ is centred in the domain, coated by an annular cloak of outer radius $r_c = 150\,\text{mm}$. The background medium is isotropic with $\lambda = \mu = 2.5\,\text{MPa}$ and $R = 2.7\,\text{kg/m}^2$ (plane stress).

The excitation is a Gaussian body-force line source at $x = 0.15\,W$ from the left physical boundary, producing approximately cylindrical pressure waves at $f = 40\,\text{kHz}$ (P-wave wavelength $\lambda_P \approx 42\,\text{mm}$, giving a void diameter of $\approx 2.4\,\lambda_P$). The frequency-domain elastodynamic equations are solved with complex-valued DOFs and Rayleigh-damping PML via a JAX-FEM finite-element solver on a triangular mesh with distance-based refinement near the cloak boundaries ($\sim$246,000 elements, $\sim$124,000 nodes).

2.2 Cloaking loss metric

We define the cloaking distortion as the relative L^2 displacement mismatch on the right physical boundary (the shadow side behind the scatterer):

$$\mathcal{D} = 100\, \frac{\|\mathbf{u}_{\text{cloak}} - \mathbf{u}_{\text{ref}}\|_{L^2(\partial\Omega_{\text{right}})}}{\|\mathbf{u}_{\text{ref}}\|_{L^2(\partial\Omega_{\text{right}})}} \quad [\%]\,, \tag{8}$$

where $\mathbf{u}_{\text{ref}}$ is the displacement in the intact background (no void) and $\mathbf{u}_{\text{cloak}}$ is the solution with void and cloak. Perfect cloaking yields $\mathcal{D} = 0$.

2.3 Results

Three configurations are compared: (i) uncoated void (void present, no cloak), (ii) Nassar cell model at varying resolutions, and (iii) continuous $\mathbf{c}_{\text{eff}}$ from (2) (the theoretical optimum for the FEM discretisation).

Table 1: Cloaking distortion $\mathcal{D}$ as a function of cell grid resolution.

Configuration	Cloak cells	$\mathcal{D}$ [%]
Uncoated void	—	70.31
Nassar 5×5	20	119.64
Nassar 10×10	68	127.11
Nassar 20×20	284	91.60
Nassar 40×40	1 124	73.74
Nassar 60×60	2 512	49.42
Nassar 80×80	4 468	45.88
Nassar 100×100	6 988	37.02
Nassar 150×150	15 716	33.77
Nassar 200×200	27 948	28.63
Continuous $\mathbf{c}_{\text{eff}}$	∞	27.51

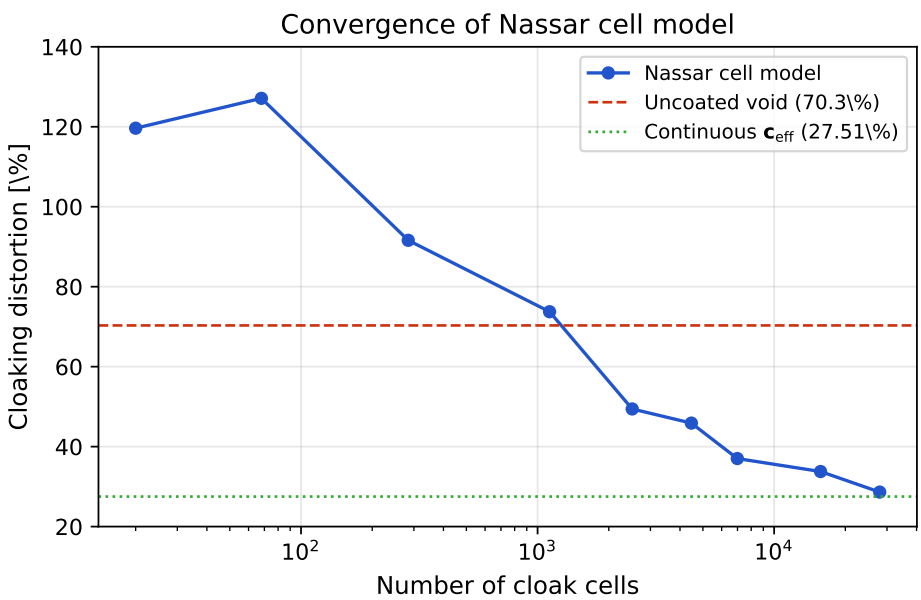

Figure 3: Convergence of the Nassar cell model (right-boundary loss). As the number of cloak cells increases, the distortion decreases from the coarse-grid value ($\mathcal{D} \approx 127\%$ at 10×10) toward the continuous limit ($\mathcal{D} \approx 27.5\%$). Above $\sim 60 \times 60$ cells the cloak outperforms the uncoated void (dashed red line), and at 200×200 cells the cell model approaches the continuous $\mathbf{c}_{\text{eff}}$ limit.

The convergence is monotonic for grids above 10×10 (Table 1, Fig. 3): the cell model begins worse than the bare void at very coarse resolution (5×5, 10×10), because the piecewise-constant approximation of the singular $\mathbf{c}_{\text{eff}}$ near r_i introduces strong impedance mismatches at cell boundaries. At $\sim 60 \times 60$ cells the cloaking quality surpasses the uncoated case, and at 200×200 cells the distortion approaches the continuous limit of 27.5%.

The residual $\sim 28\%$ distortion on the right boundary even for the continuous cloak is questionable and requires further investigation.

2.4 Mesh convergence of the continuous $\mathbf{c}_{\text{eff}}$ model

To understand the role of FEM discretisation error, the continuous $\mathbf{c}_{\text{eff}}$ cloak is solved on meshes ranging from 80×80 to 250×250 physical elements.

Table 2: Mesh convergence of the continuous $\mathbf{c}_{\text{eff}}$ model.

Mesh	$\mathcal{D}_{\text{right}}$ [%]	$\mathcal{D}_{\text{boundary}}$ [%]	$\mathcal{D}_{\text{outside}}$ [%]
80×80	41.58	4.32	4.22
120×120	31.70	2.49	2.35
160×160	27.51	1.33	1.57
200×200	27.13	1.27	1.30
250×250	23.02	1.04	1.06

A power-law fit $\mathcal{D}(h) = \mathcal{D}_\infty + A\,h^p$ to the right-boundary distortion (where $h = 1/N$ is the characteristic mesh size) yields convergence exponent $p \approx 1.35$ and asymptote $\mathcal{D}_\infty \approx 19\%$ (Fig. 4). Extrapolation predicts $\mathcal{D} \approx 21\%$ at $N = 500$ and $\mathcal{D} \approx 20\%$ at $N = 1000$, indicating that a significant portion of the $\sim 28\%$ residual distortion observed at 160×160 is FEM discretisation error rather than an intrinsic limitation of the continuous cloak.

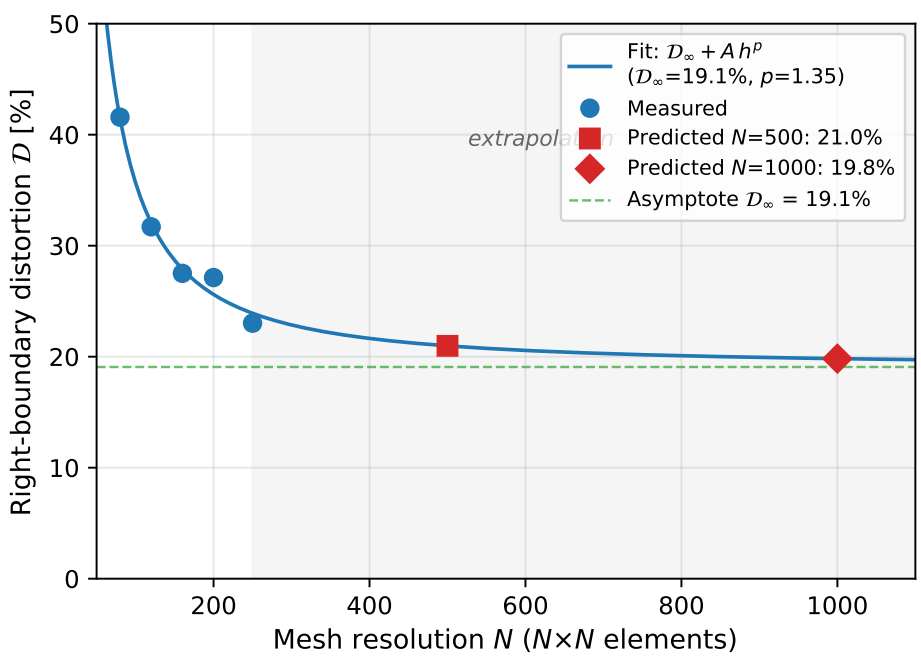

Figure 4: Mesh convergence of the continuous $\mathbf{c}_{\text{eff}}$ model (right-boundary distortion). Circles: measured data; solid line: power-law fit $\mathcal{D}_\infty + A\,h^{1.35}$ with $\mathcal{D}_\infty \approx 19\%$; diamond/square: extrapolated predictions. The grey region marks the extrapolation range beyond the finest computed mesh (250×250). Note that the fit has limited data points ($n = 5$) and the extrapolation carries substantial uncertainty ($\mathcal{D}_\infty = 19 \pm 5\%$).

2.5 Global outside-cloak distortion

In addition to the right-boundary metric $\mathcal{D}$, we evaluate the *outside-cloak distortion* $\mathcal{D}_{\text{out}}$, defined over all physical-domain nodes outside the cloak annulus:

$$\mathcal{D}_{\text{out}} = 100\,\frac{\|\mathbf{u}_{\text{cloak}} - \mathbf{u}_{\text{ref}}\|_{L^2(\Omega_{\text{phys}}\setminus\overline{\Omega}_{\text{cloak}})}}{\|\mathbf{u}_{\text{ref}}\|_{L^2(\Omega_{\text{phys}}\setminus\overline{\Omega}_{\text{cloak}})}}\ [\%]\,. \tag{9}$$

This metric captures the global wavefield perturbation due to the scatterer, rather than only the shadow-side mismatch. It is closely related to the objective function used by Wang et al. [Wan+22] in their data-driven aperiodic metamaterial cloak design, where the L^2 displacement error over the exterior domain serves as the optimisation target for inverse-designed cloaks.

Table 2 shows that $\mathcal{D}_{\text{out}}$ converges more rapidly than the right-boundary metric and is already below 1.1% at 250×250. A power-law fit $\mathcal{D}_{\text{out}}(n) = \mathcal{D}_\infty + A\, n^{-p}$ (where n is the number of outside-cloak nodes) yields $p \approx 1.37$ and $\mathcal{D}_\infty \approx 0.75\%$ (Fig. 5). Extrapolation predicts $\mathcal{D}_{\text{out}} \approx 0.80\%$ at $N = 500$ and $\mathcal{D}_{\text{out}} \approx 0.76\%$ at $N = 1000$, confirming that the continuous $\mathbf{c}_{\text{eff}}$ cloak achieves near-ideal global cloaking (sub-1% outside distortion) once the mesh is sufficiently fine.

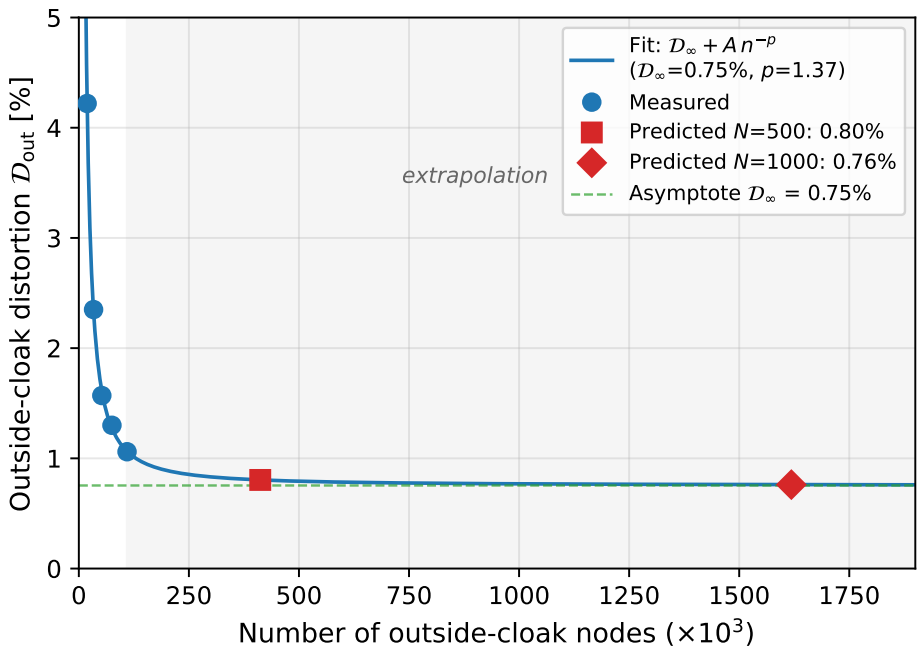

Figure 5: Mesh convergence of the outside-cloak distortion $\mathcal{D}_{\text{out}}$ for the continuous $\mathbf{c}_{\text{eff}}$ model, plotted against the number of outside-cloak nodes. Circles: measured data; solid line: power-law fit with $\mathcal{D}_\infty \approx 0.75\%$ and $p \approx 1.37$; diamond/square: extrapolated predictions. The asymptotic value $\mathcal{D}_\infty \approx 0.75\%$ serves as a reference target for the continuous cloak under this metric, comparable to the objective used in [Wan+22].

For comparison, Wang et al. [Wan+22] use the same type of metric — relative $L2$ displacement difference over the surrounding region Ω_s outside the cloak — in an elasto-static setting with a similar circular-void annular-cloak geometry ($r_c = 2\, r_i$). Their approach differs in two key respects: (i) they consider quasi-static loading (compression, pressure, shear) rather than frequency-domain wave propagation, and (ii) they work with classical Cauchy elasticity (orthotropic unit cells with four independent stiffness parameters $C_{11}, C_{12}, C_{22}, C_{33}$) rather than the Cosserat gauge with broken minor symmetry. Instead of analytical transformation-elasticity formulas, they employ a data-driven inverse design that selects aperiodic unit cells from a precomputed database. They report:

> *"With the designed cloak, the relative difference calculated from the numerical simulations is reduced from 17.2% in the voided structure to 4% in the cloaked structure, demonstrating a good cloaking performance."* — Wang et al. [Wan+22]

Under the analogous global metric $\mathcal{D}_{\text{out}}$, our continuous $\mathbf{c}_{\text{eff}}$ cloak reduces the outside distortion from 5.18% to an asymptotic $\sim 0.75\%$, substantially below the 4% achieved by Wang et al., though the comparison is not strictly apples-to-apples given the different physics.

References

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